Strength of Materials – Cheat Sheet with Examples

1. Stress and Strain

Formula: $\sigma = F / A$, $\varepsilon = \Delta L / L$
Example: A steel rod of 10 mm diameter and 1 m length subjected to 5 kN load.
• Stress $\sigma = 63.66$ MPa
• Strain $\varepsilon = 3.18 \times 10$■■
• Elongation $\Delta L \approx 0.318$ mm

2. Bending of Beams

Formula: $\sigma = (M \cdot y) / I$
Example: A simply supported wooden beam (2 m span) carries 1000 N at center.
• Maximum bending moment M = 500 Nm
• Bending stress ≈ 24 MPa

3. Torsion in Shafts

Formula: $\tau = (T \cdot r) / J$
Example: A solid shaft of 50 mm diameter under 200 Nm torque.
• Shear stress $\tau \approx 1.63$ MPa

4. Column Buckling (Euler's Formula)

Formula: $Pcr = (\pi^2 \cdot E \cdot I) / (K \cdot L)^2$
Example: A steel column of 2 m length, 40 mm diameter, pinned-pinned ends.
• Critical load Pcr ≈ 6200 N

5. Elastic Moduli Relations

$E = 2G(1 + \nu)$, $E = 3K(1 - 2\nu)$

Concept	Formula	Key Example
Stress & Strain	$\sigma = F/A$, $\varepsilon = \Delta L/L$	Steel rod elongation 0.318 mm
Bending	$\sigma = M \cdot y/I$	Beam stress ≈ 24 MPa
Torsion	$\tau = T \cdot r/J$	Shaft shear ≈ 1.63 MPa
Buckling	$Pcr = \pi^2 \cdot E \cdot I/(KL)^2$	Critical load ≈ 6200 N